

Gebze Technical University
Department of Computer Engineering
CSE 312 Spring 2026 Quiz 2 Apr 1 2026

PART A: Answer on the front side:

Consider the following set of jobs arriving at the times indicated:

Job Arrival Time CPU Time Required

A	0	2
B	0	4
C	3	1
D	3	1
E	3	1

- (a) Calculate the average waiting time under the **First-Come, First-Served (FCFS)** algorithm
- (b) Now apply the **Shortest Job First (SJF)** algorithm. Is the order you find necessarily optimal? Explain why or why not, referring to the concept of *simultaneous availability*.

PART B: Answer on the backside:

A system uses strict priority scheduling with three processes: H (high), M (medium), and L (low priority).

- (a) Describe step by step how priority inversion can occur in this scenario. What is the practical consequence observed in a real system such as the Mars Pathfinder mission?
- (b) Explain how priority donation resolves this problem. What are the two cases (**multiple donations and nested donations**) that a correct implementation must handle?